

5.4 Quartiles and outliers



In addition to the median of a single set of data, there are two related statistics called the upper and lower quartiles. If data are placed in order, then the lower quartile is central to the lower half of the data. The upper quartile is central to the upper half of the data. These quartiles are used to calculate the interquartile range, which helps to describe the spread of the data, and show whether or not any data points do not fit the rest of the data (outliers).



► Let's start: House prices

A real estate agent tells you that the median house price for a suburb in 2012 was \$753 000 and the mean was \$948 000.

- Is it possible for the median and the mean to differ by so much?
- Under what circumstances could this occur? Discuss.

- The **five-figure summary** uses the following statistical measures.

– Minimum value (Min)	the lowest value
– Lower quartile (Q_1)	the number above 25% of the ordered data
– Median (Q_2)	the middle value, above 50% of the ordered data
– Upper quartile (Q_3)	the number above 75% of the ordered data
– Maximum value (Max)	the highest value

Odd number of data values

1 2 2 3) 5 (6 6 7 9

↓ ↓ ↓

$$Q_1 = \frac{2+2}{2} = 2 \quad Q_2 = 5 \quad Q_3 = \frac{6+6}{2} = 6.5$$

Even number of data values

2 3 3 4 7 8 8 9 9

↓ ↓ ↓

$$Q_1 = 3 \quad Q_2 = 7.5 \quad Q_3 = 9$$

- Another measure of the spread of the data is the **interquartile range (IQR)**
 $\text{IQR} = \text{upper quartile} - \text{lower quartile}$
 $= Q_3 - Q_1$
- **Outliers** are data elements outside the vicinity of the rest of the data.
 A data point is an outlier if it is either:
 - less than $Q_1 - 1.5 \times \text{IQR}$ or
 - greater than $Q_3 + 1.5 \times \text{IQR}$

Five-figure summary

A set of numbers that summarise a set of data: the minimum score, first quartile, median, third quartile and maximum score

Quartiles The three values that separate the scores when a set of ordered data is divided into four equal parts

Interquartile range A measure of spread giving the difference between the upper and lower quartiles

Outlier Any value that is much larger or much smaller than the rest of the data in a set

Exercise 5D

Understanding

- 1 **a** State the five values that need to be calculated for a five-figure summary.
b Explain the difference between the range and the interquartile range.
c What is an *outlier*?
- 2 Complete the following for calculating outliers.
a Numbers below _____ $- 1.5 \times \text{IQR}$
b Numbers above $Q_3 + \text{_____} \times \text{IQR}$.
- 3 This data shows, in order, the numbers of cars owned by 10 families surveyed.
 0, 1, 1, 1, 1, 2, 2, 2, 3, 3
a Find the median (the middle value).
b By splitting the data in half, determine:
i the lower quartile Q_1 (middle of lower half)
ii the upper quartile Q_3 (middle of upper half)



- 4 For the data set with $Q_1 = 3$ and $Q_3 = 8$:
a find $\text{IQR} = Q_3 - Q_1$
b calculate $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$
c identify the name that would be given to the value 18 in the data set

Fluency

Example 9 Finding quartiles and IQR for an even number of data values

Consider this data set.

2, 2, 4, 5, 6, 8, 10, 13, 16, 20

- a** Find the upper quartile (Q_3) and the lower quartile (Q_1).
- b** Determine the IQR.

Solution

a

2	2	4	5	6	8	10	13	16	20
		↑			↑				
		Q_1			Q_3				

$Q_1 = 4$ and $Q_3 = 13$

$$Q_2 = \frac{6 + 8}{2} = 7$$

Explanation

The data is already ordered. Since there is an even number of values, split the data in half to locate the median.

Q_1 is the middle value of the lower half:

2 2 4 5 6

Q_3 is the middle value of the upper half.

b $\text{IQR} = 13 - 4$
 $= 9$

$\text{IQR} = Q_3 - Q_1.$

- 5 For these data sets, find:
- the upper quartile (Q_3) and the lower quartile (Q_1)
 - the IQR
- 3, 4, 6, 8, 8, 10
 - 10, 10, 11, 14, 14, 15, 16, 18, 20, 21
 - 41, 49, 53, 58, 59, 62, 62, 65
 - 1.2, 1.7, 1.9, 2.2, 2.4, 2.5, 2.9, 3.2

For an even number of data values, split the ordered data in half:

2 4 7 | 8 10 12

↑ ↑

Q_1 Q_3

$IQR = Q_3 - Q_1$

Example 10 Finding quartiles and IQR for an odd number of data values

Consider this data set.

2.2, 1.6, 3.0, 2.7, 1.8, 3.6, 3.9, 2.8, 3.8

- Find the upper quartile (Q_3) and the lower quartile (Q_1).
- Determine the IQR.

Solution

a

1.6	1.8		2.2	2.7	2.8	3.0	3.6		3.8	3.9
					↑					
$Q_1 = \frac{1.8 + 2.2}{2}$						$Q_3 = \frac{3.6 + 3.8}{2}$				
$= \frac{4.0}{2}$						$= \frac{7.4}{2}$				
$= 2.0$						$= 3.7$				

Explanation

First order the data and locate the median (Q_2).

Split the data in half, i.e. either side of the median.

Q_1 is the middle value of the lower half; for two middle values, average the two numbers.

Q_3 is the middle value of the upper half.

$$IQR = Q_3 - Q_1$$

b $IQR = 3.7 - 2.0$
 $= 1.7$

- 6 For these data sets, find:
- the upper quartile (Q_3) and the lower quartile (Q_1)
 - the IQR
- 1, 2, 4, 8, 10, 11, 14
 - 10, 7, 14, 2, 5, 8, 3, 9, 2, 12, 1
 - 0.9, 1.3, 1.1, 1.2, 1.7, 1.5, 1.9, 1.1, 0.8
 - 21, 7, 15, 9, 18, 16, 24, 33, 4, 12, 13, 18, 24

For an odd number of data values, split ordered data in half, leaving out the middle value.

0 2 4 7 9 14 16

↑ ↑

Q_1 Q_3

Example 11 Finding the five-figure summary and outliers

The following data set represents the number of flying geese spotted on each day of a 13-day tour of England.

5, 1, 2, 6, 3, 3, 18, 4, 4, 1, 7, 2, 4

- For the data, find:
 - the minimum and maximum number of geese spotted
 - the median
 - the upper and lower quartiles
 - the IQR
- Find any outliers
- Can you give a possible reason for why the outlier occurred?

Solution

- a i** Min = 1, max = 18
ii 1, 1, 2, 2, 3, 3, 4, 4, 4, 5, 6, 7, 18
 $\therefore$ median = 4
iii Lower quartile (Q_1) = $\frac{2+2}{2}$
 $= 2$
Upper quartile (Q_3) = $\frac{5+6}{2}$
 $= 5.5$
iv IQR = $5.5 - 2$
 $= 3.5$
b $Q_1 - 1.5 \times \text{IQR} = 2 - 1.5 \times 3.5$
 $= 2 - 5.25$
 $= -3.25$
 $Q_3 + 1.5 \times \text{IQR} = 5.5 + 1.5 \times 3.5$
 $= 5.5 + 5.25$
 $= 10.75$
 $\therefore$ the outlier is 18.
c Perhaps a flock of geese was spotted that day.

Explanation

Look for the largest and smallest numbers and order the data:

1 1 2 | 2 3 3 | 4 | 4 4 5 | 6 7 18
 $\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $Q_1 \quad \quad Q_2 \quad \quad Q_3$

Since Q_2 falls on a data value, it is not included in the lower or higher halves when Q_1 and Q_3 are calculated.

$$\text{IQR} = Q_3 - Q_1$$

A data point is an outlier if it is less than $Q_1 - 1.5 \times \text{IQR}$ or greater than $Q_3 + 1.5 \times \text{IQR}$.

There are no numbers less than -3.25 but 18 is greater than 10.75.

- 7** The following numbers of cars were counted on each day for 15 days, travelling on a quiet suburban street.

10, 9, 15, 14, 10, 17, 15, 0, 12, 14, 8, 15, 15, 11, 13

- a** For the given data, find:
i the minimum and maximum number of cars counted
ii the median
iii the lower and upper quartiles (Q_1 and Q_3)
iv the IQR
b Find any outliers.
c Give a possible reason for the outlier.

- 8** Summarise the data sets below by finding the:

- i** minimum and maximum values **ii** median (Q_2)
iii lower and upper quartiles (Q_1 and Q_3) **iv** IQR
v any outliers
a 4, 5, 10, 7, 5, 14, 8, 5, 9, 9
b 24, 21, 23, 18, 25, 29, 31, 16, 26, 25, 27
c 10, 13, 2, 11, 10, 8, 24, 12, 13, 15, 12
d 3, 6, 10, 11, 17, 4, 4, 1, 8, 4, 10, 8

Outliers:
more than $Q_3 + 1.5 \times \text{IQR}$
or
less than $Q_1 - 1.5 \times \text{IQR}$

